

Quiz 1

Show all your work clearly. Write precisely. **Justify everything.** No credit for unsupported answers.

1. Consider the matrix $A \in \mathbb{R}^{3 \times 4}$ given by $A = \begin{pmatrix} 0 & 0 & 2 & 4 \\ 1 & 5 & 2 & 1 \\ 1 & 5 & 3 & 3 \end{pmatrix}$. Show that the Reduced Row Echelon Form (RREF) of A is $R = \begin{pmatrix} 1 & 5 & 0 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. Clearly specify the operation at each step.
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2. Solve the linear system $A\mathbf{x} = \mathbf{b}$, where A is the matrix from Problem 1 and $\mathbf{b} = \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$. Describe the complete solution set S precisely. Then express S as a “translate of a subspace”, meaning you must specify a subspace W and a (constant) vector $\mathbf{p}$ such that $S = W + \mathbf{p}$, i.e., $S = \{\mathbf{w} + \mathbf{p} \mid \mathbf{w} \in W\}$.
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3. For general $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \in \mathbb{R}^3$, find necessary and sufficient condition(s) on b_1, b_2, b_3 for the linear system $A\mathbf{x} = \mathbf{b}$ to be consistent, where A is the matrix from Problem 1.
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4. Let U be the subspace of $\mathbb{R}^4$ spanned by the vectors $\mathbf{v}_1 = \begin{pmatrix} -12 \\ 1 \\ 14 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 15 \\ 0 \\ -23 \\ 1 \end{pmatrix}$. Find a matrix M , with as few rows as possible, such that U is precisely the null space of M (i.e., $U = \{\mathbf{x} \in \mathbb{R}^4 \mid M\mathbf{x} = \mathbf{0}\}$).

Hint: Look carefully at the given vectors and try to exploit some very helpful slots! (Note: You may realize that this problem is an instance of Problem 2b in Tutorial 2, which asked you for a method to find, for any given set of vectors in $\mathbb{R}^n$, such a “magic matrix”. That problem was stated as optional, and you do NOT need it here if you follow the hint.)

5. Let M be a $p \times q$ matrix with column vectors $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_q \in \mathbb{R}^p$ and row vectors $\mathbf{r}_1^T, \mathbf{r}_2^T, \dots, \mathbf{r}_p^T \in \mathbb{R}^q$. Prove the following from the definitions of span and linear independence.

- (a) Show that if $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_q$ are linearly independent, then $\text{span}(\mathbf{r}_1^T, \mathbf{r}_2^T, \dots, \mathbf{r}_p^T) = \mathbb{R}^q$.
- (b) Show that if $\text{span}(\mathbf{r}_1^T, \mathbf{r}_2^T, \dots, \mathbf{r}_p^T) = \mathbb{R}^q$ then $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_q$ are linearly independent.
- (c) So (the *rows* of any matrix are linearly independent) $\Leftrightarrow$ (its *columns* span the ambient space). How?

You may not directly appeal to any part of our table of equivalences, but you can of course recreate the logic that went into proving that table. You can freely use facts we developed before the table (in particular about solving a linear system of equations) but in each instance clearly state which fact you are using and how. Try to find the shortest path involving as few notions as possible. **The word *dimension* is not allowed**, so of course you may not use its invariance, even implicitly.